

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	15 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture:

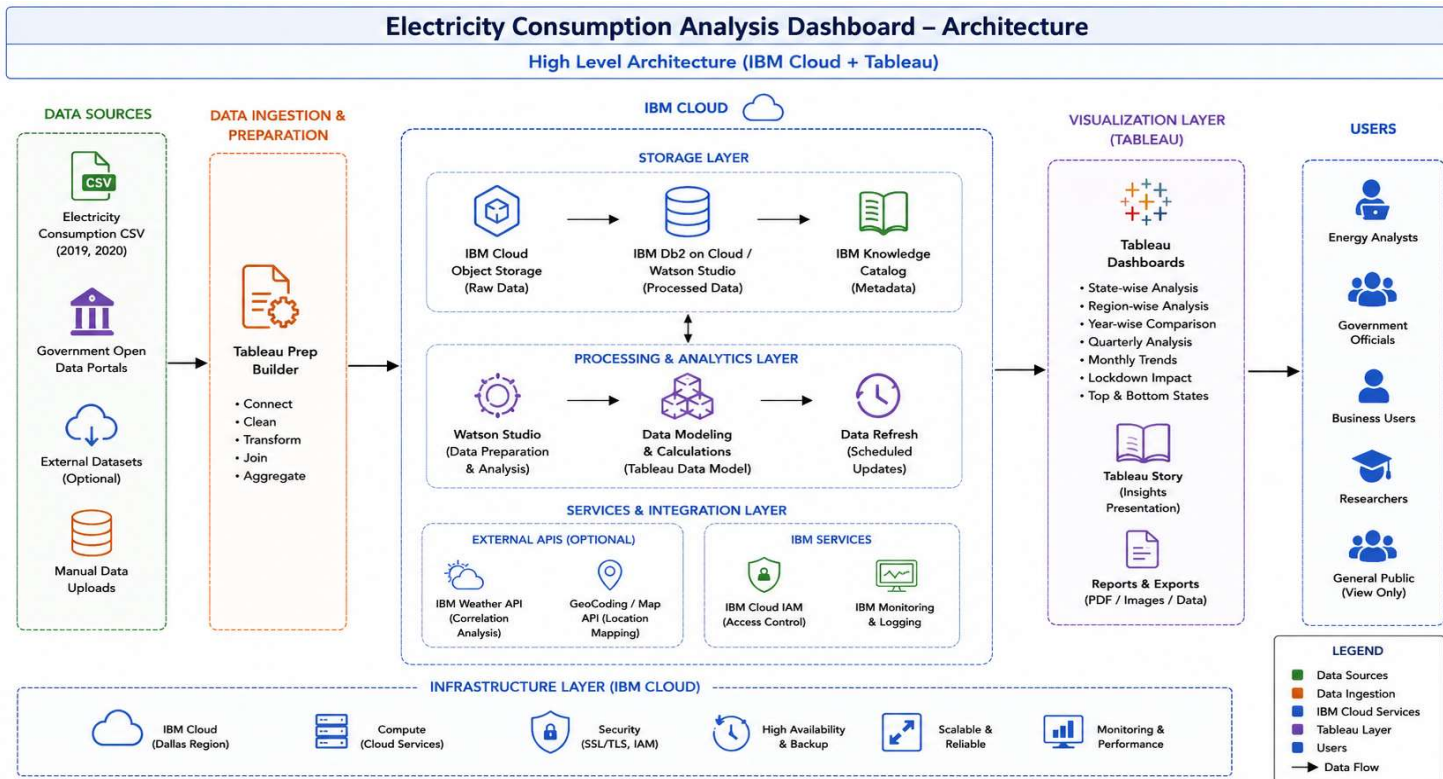


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards, worksheets, and story for visualizing electricity consumption trends	Tableau Desktop / Tableau Public
2	Data Source	State-wise electricity consumption dataset (2019–2020)	CSV Dataset
3	Data Preparation	Import, clean, transform, and prepare the dataset for analysis	Tableau Desktop (Data Source & Data Interpreter)
4	Data Processing	Create calculated fields, parameters, filters, and aggregations	Tableau Calculations, LOD Expressions, Parameters
5	Visualization Layer	Develop interactive charts, maps, dashboards, and story points	Tableau Desktop
6	Database / Data Storage	Store extracted data for analysis	Tableau Hyper Extract (.hyper)
7	Mapping Service	Display geographical distribution of electricity consumption	Tableau Maps (Mapbox / OpenStreetMap)
8	Analytics Engine	Analyze state-wise, region-wise, quarterly, monthly, and yearly consumption	Tableau Analytics
9	Reporting	Publish dashboards and generate interactive reports	Tableau Public
10	File Storage	Store Tableau workbook, packaged workbook, and dataset	.twb, .twbx, CSV Files
11	Deployment Platform	Host dashboards online for public viewing	Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks and tools used in the project.	Tableau Public, OpenStreetMap (Mapbox), CSV Dataset
2	Security Implementation	Public dashboard provides read-only access, while editing is restricted to the project owner.	Tableau Public Permissions, Tableau Desktop
3	Scalable Architecture	Dashboard supports adding new years, states, regions, or updated datasets without redesigning the entire system.	Tableau Data Model, Hyper Extract
4	Availability	Dashboard is available online through Tableau Public and can be accessed anytime using a web browser.	Tableau Public Cloud
5	Performance	Optimized data extracts and visualizations ensure quick loading and smooth interaction.	Tableau Hyper Extract, Tableau Engine
6	Data Accuracy	Data is validated during import and visualizations accurately represent electricity consumption values.	Tableau Data Source, Calculated Fields
7	Usability	Interactive filters, dashboards, maps, and story points make analysis simple for users.	Tableau Dashboard, Filters, Story
8	Compatibility	Dashboard can be viewed on Windows, macOS, and modern web browsers.	Tableau Desktop, Tableau Public
9	Maintainability	New datasets can be imported and the workbook can be refreshed without rebuilding the dashboards.	Tableau Refresh, Data Source Management

S.No	Characteristics	Description	Technology
10	Reliability	Dashboard consistently displays correct charts and maps after data refresh and publishing.	Tableau Desktop, Tableau Public
11	Portability	Workbook can be shared as .twbx and published through Tableau Public using a shareable URL.	Tableau Packaged Workbook (.twbx), Tableau Public

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>